

4(a)	time for one oscillation/vibration/cycle or time between <u>adjacent</u> wavefronts (passing the same point) or <u>shortest</u> time between two wavefronts (passing the same point)	B1
4(b)	(when two or more) waves meet/overlap (at a point)	B1
	(resultant) displacement is sum of the individual displacements	B1
4(c)(i)	microwave(s)	B1
4(c)(ii)	$v = \lambda / T$ or $v = f\lambda$ and $f = 1/T$	C1
	$T = 0.040 / 3.00 \times 10^8$	C1
	$= 1.33 \times 10^{-10} \text{ (s)}$ $= 1.33 \times 10^{-10} / 10^{-12} \text{ (ps)}$ $= 130 \text{ ps}$	A1
4(c)(iii)	$(1.380 - 1.240) / 0.040 = 3.5$ or $1.380 / 0.040 - 1.240 / 0.040 = 3.5$	A1
4(c)(iv)	phase difference = 1260° or 180°	A1
4(c)(v)	(always) zero	A1
4(c)(vi)	increase in distance between (adjacent intensity) maxima/minima	A1